

AI Snake Game

Prompt Engineering Report

Project: AI Snake Game (React + Vite + Tailwind CSS) | Tool: Claude Code
Development Period: June 14 – June 16, 2026

1. Introduction

This project is a Snake Game made using React. The whole game was built by giving instructions (called "prompts") to an AI coding tool named Claude Code. Instead of writing every line of code by hand, the game was created step by step by telling the AI what to build and what to fix. This report explains what prompts were used, what worked, what did not work, and what was learned.

2. Prompts Used During Development

In total, 3 prompts were given to build and improve the game.

Step 1: Building the Game

Prompt given:

"Create a complete Snake Game. Using react. Requirements: Single page application, Modern UI, Responsive design, Score system, Game over screen, Restart button. Provide complete project structure. make sure code will be beginner friendly"

What happened: The AI created the entire project from scratch — all the files, folders, the game logic (movement, score, collision), and the basic design. It also installed everything needed and started the game to make sure it worked.

Step 2: Improving the Design

Prompt given:

"use tailwind not simple css for styling and make sure code will be beginner friendly. design will be fully responsive and and every function workly perfectly"

What happened: The AI replaced the old plain CSS styling with Tailwind CSS (a modern styling tool), making the game look better and work properly on both mobile and desktop screens. While doing this, it also found and fixed a hidden bug that was affecting how the game tracked the food.

Step 3: Cleaning Up the Code

Prompt given:

"use short comments not long"

What happened: The AI went through every file and shortened all the comments, removing long explanations and keeping only short, useful notes. No functionality was changed — this step was only about making the code neater and easier to read.

3. What Worked Well

- One detailed prompt at the start was enough to build the whole working game in one go.
- Short, simple follow-up prompts (just one line) were enough to make big changes across all the files at once.
- The AI checked its own work — it ran the game after every change to make sure it actually worked, instead of just assuming the code was correct.
- While fixing the design, the AI also noticed and fixed a bug on its own, even though it wasn't asked to.

4. What Did Not Work / Problems Faced

- **Windows command problem:** One command used to install the project files failed because of a Windows-specific issue, and had to be run in a different way to work.
- **Slow installation:** Installing the project's required packages took a long time and did not show any progress on screen, so it had to be checked again and again to know when it was finished.
- **No step-by-step saving:** All the changes were saved to GitHub as just one single update, instead of saving each step separately. This made it harder to look back and see exactly what changed at each stage.
- **Missing files:** After moving the project to a new folder for submission, some of the game's code files were missing from the folder (though they were still saved in the project's history).

5. How the Prompts Were Improved Over Time

Stage	Type of Prompt	What It Covered
1st Prompt	Long and detailed, listing every requirement	Built the entire game from nothing
2nd Prompt	Short, but asked for more than one thing at once	Improved design and responsiveness
3rd Prompt	Very short, asked for only one thing	Cleaned up code comments

The first prompt had to be long because there was no code yet — everything had to be explained. Once the game existed, later prompts could be much shorter, because the AI already understood the project and only needed to know what small thing to change next.

6. Conclusion

Building this Snake Game showed that AI tools like Claude Code can build a complete, working project from a clear and detailed first instruction, and then can be improved easily with short follow-up instructions. The main problems faced were small technical issues (like Windows command differences and slow installs), not problems with the prompts themselves. Overall, the step-by-step prompting approach worked well and produced a complete, modern, and responsive Snake Game.